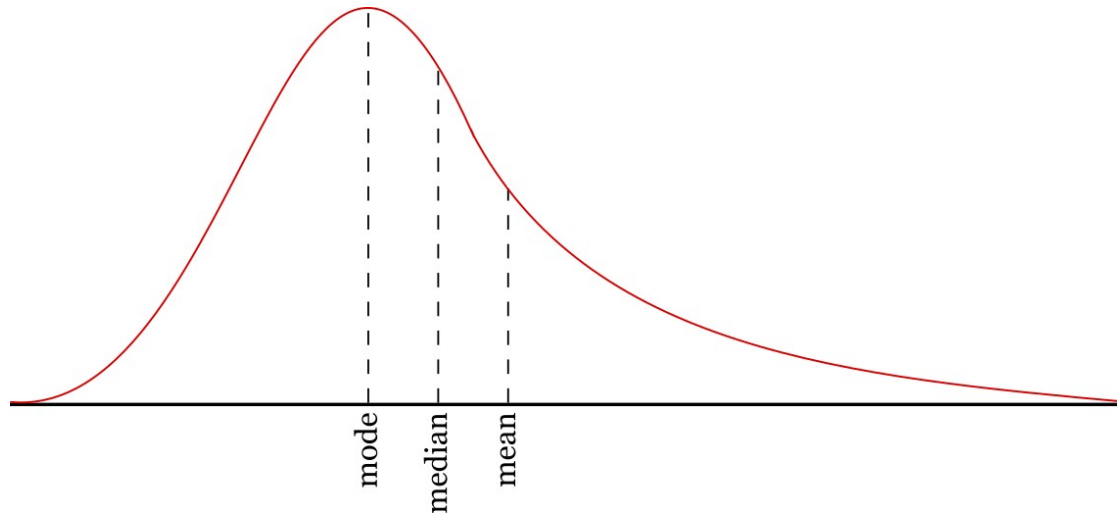


Lecture 8

Descriptive Statistics



Announcements

- ❖ Lab 2 today
 - First time getting your hands on a dataset!

Last Few Times

- ❖ Using Python and Pandas
 - Loading data from csv
 - Filtering rows and columns
- ❖ GitHub for version control, storage, and distribution of projects

Today

- ❖ What is a statistic?
- ❖ Central tendency
- ❖ Spread and variation

What is a statistic?

- ❖ A summary of the data
- ❖ Often captures one aspect or feature of a dataset

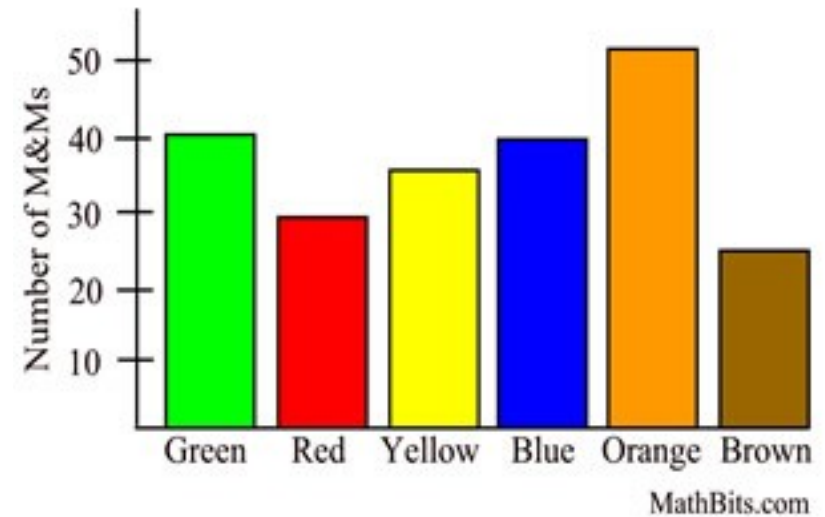
What is a statistic?

- ❖ A summary of the data
- ❖ Often captures one aspect or feature of a dataset
- ❖ We often want to describe a dataset with a small number of statistics
- ❖ Numerical summaries
 - Averages, counts, quantiles, measures of spread, correlation coefficients
- ❖ Visualizations
 - Histograms, scatterplots, pie charts

Counts & Histograms

How many of each value are in our dataset?

- ❖ Gives us a full summary of the variable in our dataset

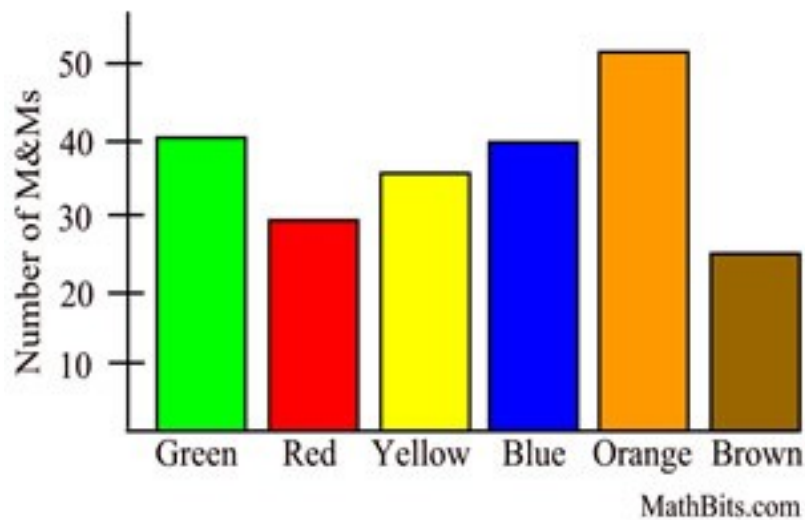


Number of Colors in Bag of M&M Candies

Counts & Histograms

How many of each value are in our dataset?

- ❖ Gives us a full summary of the variable in our dataset
- ❖ How might this go wrong?
 - Think about a numeric variable

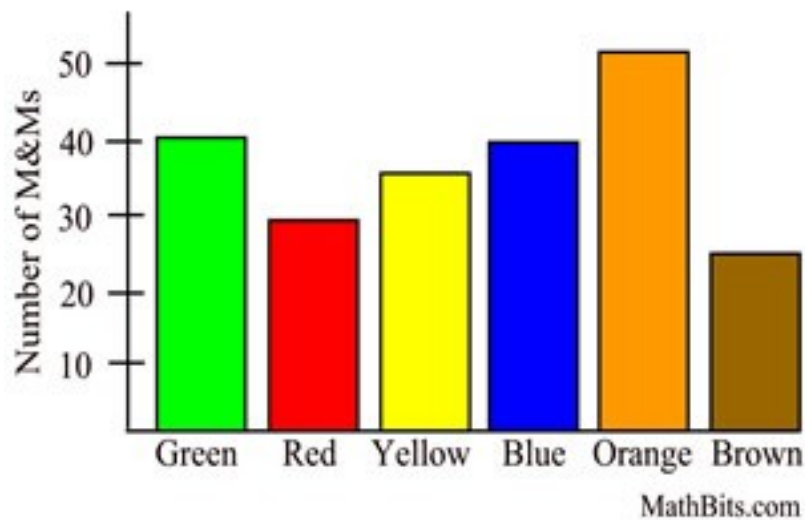


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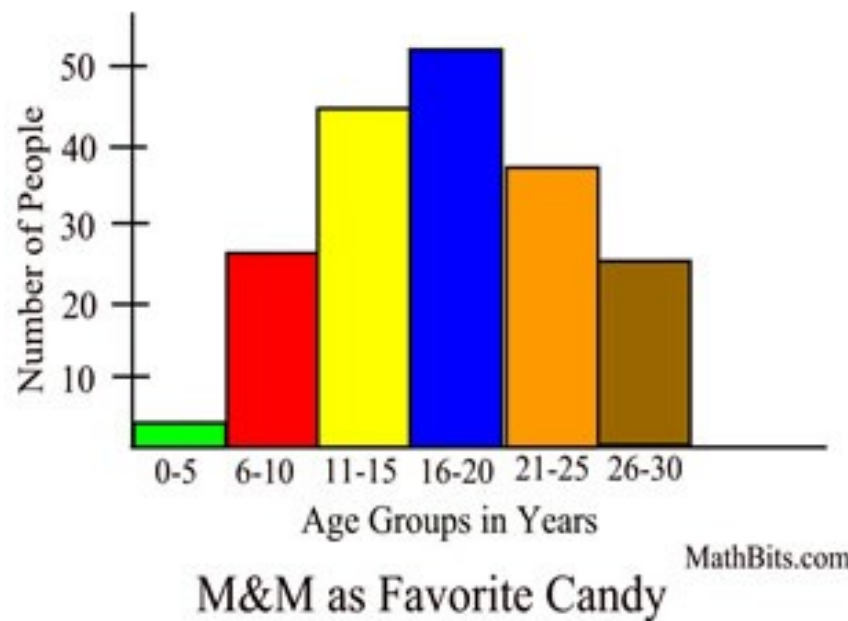


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Counts & Histograms

How many of each value are in our dataset?

- ❖ Gives us a full summary of the variable in our dataset
- ❖ How might this go wrong?
 - Think about a numeric variable
 - It's possible that every value is unique!
- ❖ For numeric variables, we tend to "bin" the data into "buckets"
 - Think of this as creating a new variable that's easier to visualize from the existing one



Centrality

- ❖ For variables that can be ordered, it's helpful to know a typical value

$$x = [x_1, x_2, x_3, \dots, x_n]$$

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- ❖ Mean (or average)
 - Sum all the values and divide by n ; denoted by $\bar{x}$
- ❖ Median
 - Sort the values and find the middle value (or average the two middle values)

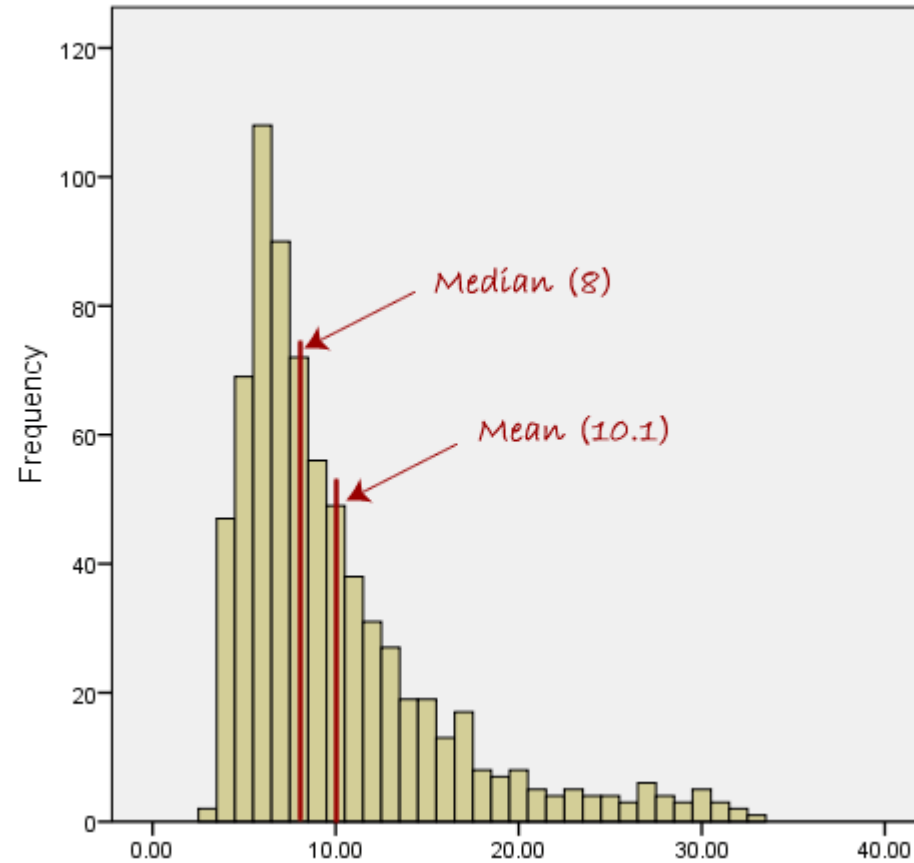
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 - Sort the values and find the middle value (or average the two middle values)
- ❖ Why do we need two measures of centrality?
 - Skew: our variable distribution may be "lopsided"

Centrality



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 - 50% of the dataset has a value less than or equal to the median

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- ❖ A quantile generalizes the median to any %
 - Sometimes called a percentile as well
- ❖ Multiple quantiles can summarize a variable distribution
 - Common quantiles: 0%, 25%, 50%, 75%, 100%

Position

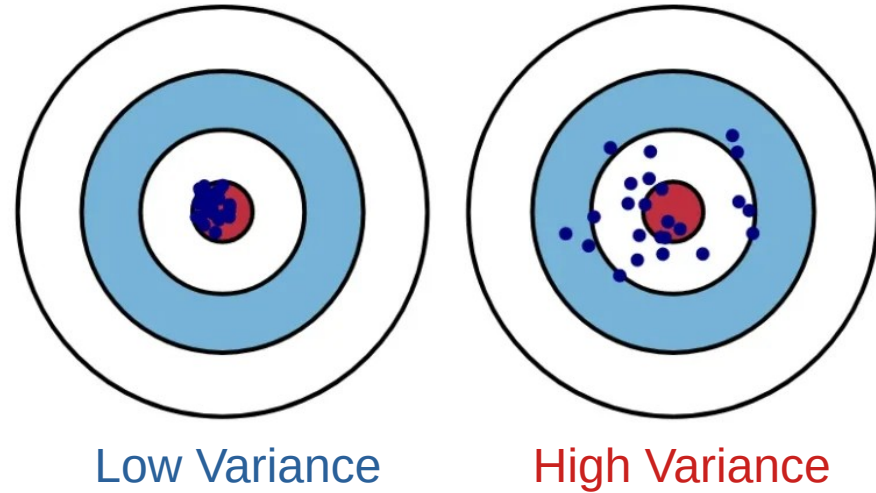
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0% quantile is min

100% quantile is max

Spread

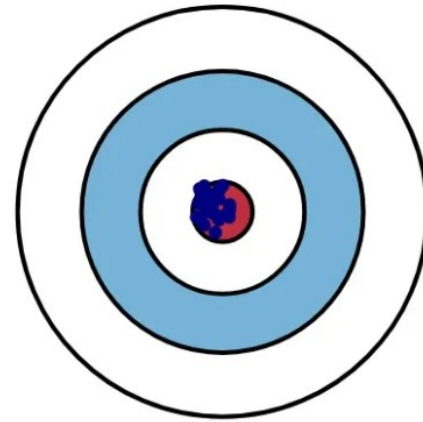
- ❖ Variance is a measure of how far the values in the dataset are from the mean



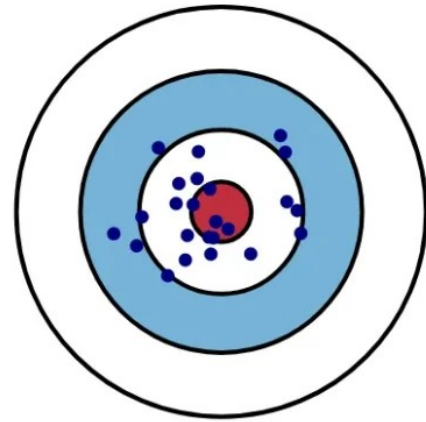
Spread

- ❖ Variance is a measure of how far the values in the dataset are from the mean
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$$Var(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$



Low Variance



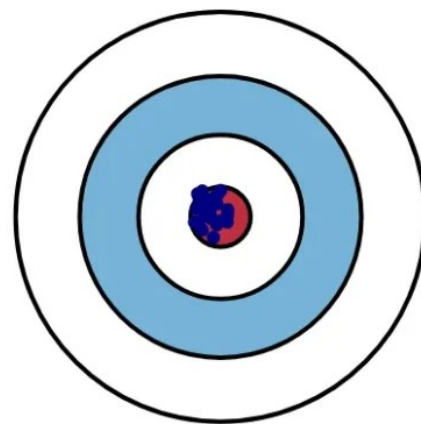
High Variance

Spread

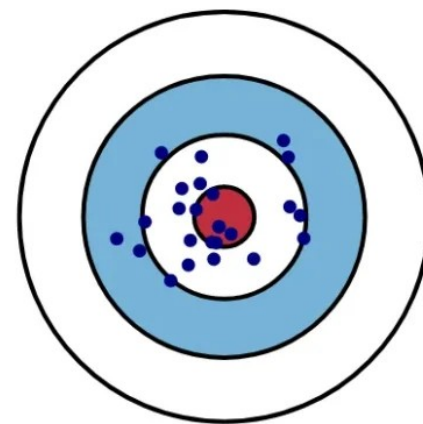
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$$\text{Var}(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

- ❖ Standard deviation is the square root of the variance
 - Same units as the variable
 - Typical distance of value from mean



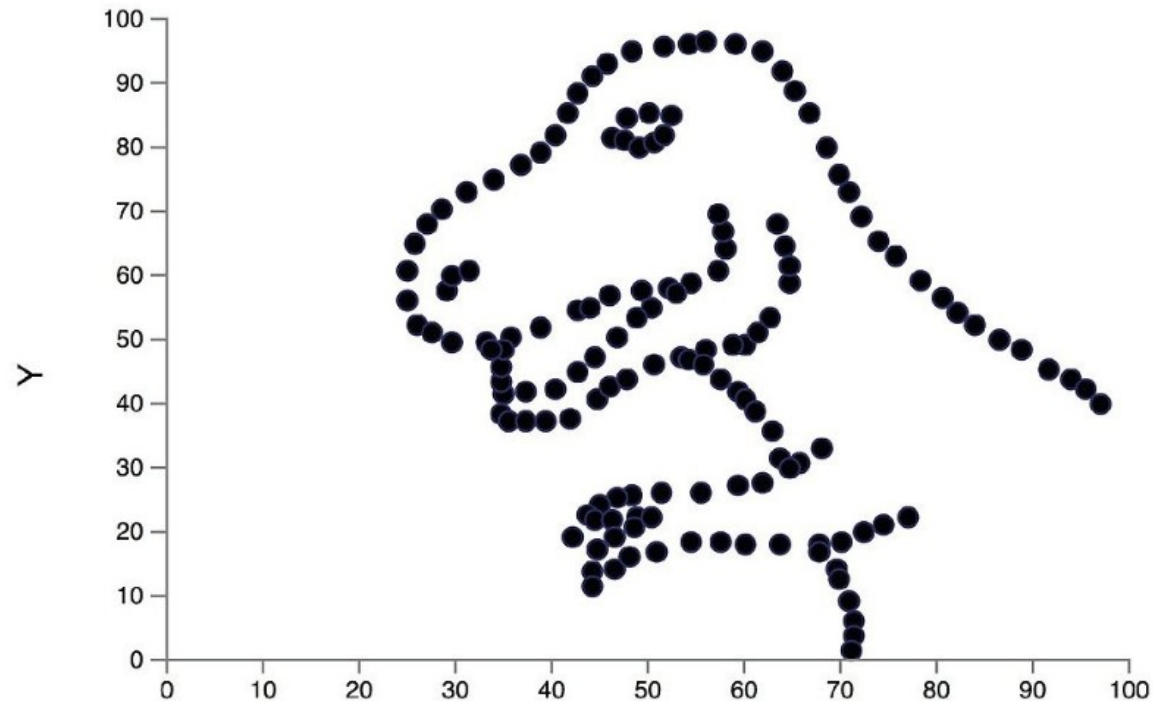
Low Variance



High Variance

Limitations

$N = 146$; $X \text{ mean} = 53.7355$; $X \text{ SD} = 15.7801$; $Y \text{ mean} = 49.0995$; $Y \text{ SD} = 24.0605$; Pearson correlation = -0.143



Next Time

- ❖ Read DS Ch. 3
- ❖ Finish up descriptive statistics
- ❖ Start on data wrangling